

UX-0007 — Butlerly Design System Foundations

Version 1.0

Purpose

This document defines the visual, interaction, content, and component foundations for Butlerly Version 1. It establishes the shared design language used across smartphone, tablet, and desktop while preserving native platform behavior. It governs tokens, typography, color, spacing, elevation, icons, controls, navigation, data presentation, AI states, motion, accessibility, theming, content, and design-system governance.

1. Design System Objective

The Butlerly design system must make the product feel calm, trustworthy, private, efficient, and understandable. It should reduce visual and cognitive overhead while helping users act confidently on sensitive personal information.

The system must:

- Preserve one recognizable Butlerly identity across three native experiences.
- Prefer platform-native conventions over custom behavior when they improve familiarity or accessibility.
- Make privacy, AI interpretation, synchronization, and system state understandable without adding noise.
- Support local-first and offline behavior as normal product states.
- Scale from fast one-handed capture to large-screen review and analysis.
- Remain usable with large text, screen readers, keyboard, pointer, touch, reduced motion, and high contrast.
- Be token-driven so design and engineering share the same source of truth.

2. Design Principles

2.1 Trust Before Delight

Visual polish must never obscure consequences, privacy boundaries, uncertainty, or recovery. Trust is created through clarity, consistency, visible control, and reliable behavior.

2.2 Quiet by Default

Routine processing, synchronization, and successful background work should remain visually quiet. Attention is reserved for meaningful exceptions, security changes, degraded states, and decisions that require the user.

2.3 One Clear Primary Action

Each screen, card, dialog, and review state should identify one primary action. Secondary actions remain available but visually subordinate.

2.4 Progressive Disclosure

Show the minimum information needed to make the current decision. Reveal source evidence, history, diagnostics, and advanced controls on demand.

2.5 Native, Not Identical

Smartphone, tablet, and desktop share the same hierarchy, terminology, states, and tokens, but use native controls, navigation, spacing, and window behavior appropriate to each platform.

2.6 Reversible by Design

Whenever an action can be reversed, the interface should favor Undo and version history over repeated warning dialogs.

2.7 Evidence Near Interpretation

When Butlerly shows an AI-generated value or recommendation, the supporting source and explanation must be accessible near the result.

3. System Architecture

The design system is organized into five layers:

Layer 1 — Foundations

Color, typography, spacing, shape, elevation, motion, iconography, and content rules.

Layer 2 — Primitives

Text, icons, dividers, surfaces, badges, avatars, progress indicators, and layout containers.

Layer 3 — Controls

Buttons, text fields, search fields, toggles, checkboxes, radios, segmented controls, date controls, menus, lists, tables, and disclosure controls.

Layer 4 — Patterns

Navigation, onboarding, capture, import, review, search, AI explanation, consent, device management, synchronization, offline, error, undo, and destructive-action patterns.

Layer 5 — Product Compositions

Home, Inbox, Records, Assistant, Settings, record detail, import center, device setup, and other screen-level assemblies.

4. Token Model

Token naming follows:

category.role.variant.state

Examples:

- color.text.primary
- color.surface.elevated
- spacing.component.medium
- radius.control.standard
- motion.duration.fast

- typography.body.standard

Tokens are semantic. Product code must not use raw color values, fixed font names, or arbitrary spacing where a token exists.

Token groups:

- color
- typography
- spacing
- sizing
- radius
- border
- elevation
- opacity
- motion
- icon
- layout
- z-order

Platform implementation may map tokens to native equivalents while preserving semantic meaning.

5. Color System

5.1 Color Roles

The system uses semantic roles rather than brand-color decoration.

Core roles:

- Background — application canvas.
- Surface — cards, panels, sheets, and contained areas.
- Elevated Surface — dialogs, menus, popovers, and temporary overlays.
- Text Primary — primary readable content.
- Text Secondary — supporting information.
- Text Tertiary — metadata and low-emphasis content.
- Border — separation and control outlines.
- Accent — primary interactive emphasis.
- Success — completed or healthy state.
- Warning — attention without failure.
- Error — failure or invalid state.
- Security — device, authentication, privacy, or trust event.
- AI — AI-generated, inferred, or recommended content when explicit distinction is useful.
- Selection — selected list, row, or control state.

5.2 Color Rules

- Color is never the only signal.

- Financial values must not rely on red/green alone.
- Confidence uses labels plus icons or structure, not color alone.
- Accent color should identify interaction, not decorate every screen.
- Large background areas use neutral tones to reduce fatigue.
- Security and destructive colors remain reserved for their semantic purpose.
- Dark mode is designed independently rather than created through simple inversion.

5.3 Contrast

All text, icons, controls, focus indicators, and meaningful graphics must meet the current Butlerly accessibility standard. When platform guidance exceeds the Butlerly minimum, the stricter requirement applies.

6. Typography

6.1 Font Strategy

Use the native system font on each platform by default. This preserves readability, localization, Dynamic Type or equivalent scaling, accessibility, performance, and platform familiarity.

6.2 Type Roles

- Display — rare use for onboarding or major empty-state promise.
- Title Large — primary page title.
- Title — section or major card title.
- Heading — subsection title.
- Body — primary reading text.
- Body Strong — emphasized body text.
- Callout — short supportive guidance.
- Label — control and compact metadata label.
- Caption — secondary metadata, timestamps, provenance.
- Monospaced Data — identifiers, hashes, diagnostics, or fixed-width comparisons only.

6.3 Typography Rules

- Do not use font weight alone to express hierarchy.
- Support text scaling without clipping or overlapping.
- Avoid all caps except short platform-standard labels where accessibility remains acceptable.
- Use tabular numerals for aligned amounts, dates, and tables when supported.
- Long financial values wrap or adapt without truncating critical digits.
- Headings remain concise and action-oriented.

7. Spacing and Layout

7.1 Base Scale

Use a 4-point base spacing scale with semantic tokens.

Recommended increments:

- 4 — micro spacing.
- 8 — compact spacing.

- 12 — small component spacing.
- 16 — standard content spacing.
- 24 — section spacing.
- 32 — large grouping spacing.
- 48 — major separation.
- 64 — large-screen structural spacing.

7.2 Layout Rules

- Related content is closer than unrelated content.
- Dense desktop layouts must still preserve readable grouping.
- Smartphone screens prioritize one primary column.
- Tablet uses adaptive split views when they improve continuity.
- Desktop uses persistent navigation and multi-pane layouts only when each pane has a clear role.
- Important content must not depend on hover.
- Safe areas, window chrome, keyboard, and platform insets are respected.

7.3 Content Width

Long-form explanations and settings text should use a readable maximum line length. Data tables and comparison views may expand when additional width improves comprehension.

8. Shape, Borders, and Elevation

8.1 Radius

Use a restrained radius system:

- Small — tags, compact controls, thumbnails.
- Standard — buttons, inputs, cards.
- Large — sheets, major cards, onboarding panels.
- Full — circular icon buttons, avatars, status dots.

8.2 Borders

Borders define structure, focus, selection, and validation. Avoid excessive card outlines that make every region compete for attention.

8.3 Elevation

Elevation is used for temporary layering, not decoration.

Levels:

- Base — normal content.
- Raised — card or floating toolbar where needed.
- Overlay — menu, popover, sheet.
- Modal — dialog requiring focused attention.

Dark mode requires adjusted shadows and surface contrast rather than stronger opacity alone.

9. Iconography

- Prefer platform-native symbols where meaning is established.
- Use a consistent stroke weight and optical size.
- Every icon-only control requires an accessible name and tooltip where pointer use is expected.
- Do not create different icons for the same action across screens.
- Avoid decorative AI sparkle icons as the sole identifier of AI behavior.
- Security, synchronization, offline, warning, and error icons must be visually distinct.
- Icons accompany, not replace, critical text in unfamiliar or consequential actions.

10. Motion and Feedback

10.1 Motion Principles

Motion communicates relationship, progression, and result. It must never imply that work is complete before system confirmation.

10.2 Motion Categories

- Immediate feedback — press, hover, selection, focus.
- Navigation transition — hierarchy or pane change.
- Progress transition — known or indeterminate work.
- State transition — saved, merged, restored, revoked.
- Attention transition — warning or new review item.

10.3 Rules

- Prefer short, subtle transitions.
- Avoid decorative looping animations.
- Processing animation must not conceal long waits.
- Reduced-motion settings remove nonessential movement and replace spatial transitions with fades or immediate changes.
- Haptics may confirm capture, save, error, or destructive confirmation where platform-appropriate.
- Sound is never required to understand state.

11. Interaction States

Every interactive component must define:

- Default.
- Hover where applicable.
- Focus.
- Pressed.
- Selected.
- Disabled.
- Loading.
- Success where applicable.
- Warning where applicable.
- Error.
- Read-only.

Focus indicators must remain visible in light mode, dark mode, high contrast, and on selected surfaces.

12. Button System

12.1 Button Hierarchy

Primary

One per decision area. Used for the recommended action.

Secondary

Alternative action with less emphasis.

Tertiary

Low-emphasis action, often text or icon plus text.

Destructive

Used only for destructive actions. Confirmation and authentication rules come from UX-0004 and UX-0005.

Icon Button

Used for common, compact actions with accessible naming.

12.2 Button Rules

- Labels begin with a verb.
- Avoid generic labels such as OK when a specific result is known.
- Loading state preserves button width and prevents duplicate submission.
- Disabled controls require an understandable reason when the reason is not obvious.
- Destructive styling is not used for reversible removal from a view.

13. Input and Form System

13.1 Input Types

- Text.
- Numeric/amount.
- Date and time.
- Search.
- Select/menu.
- Segmented selection.
- Toggle.
- Checkbox.
- Radio group.
- File input.
- Voice input where supported.

13.2 Form Rules

- Use visible labels; placeholder text is not a label.
- Explain expected format before error where practical.
- Validate at the right time: not prematurely, not only at final submission.
- Preserve user input after failure.
- Display source evidence beside uncertain extracted values.
- AI proposals are visually distinguishable but editable through the same control.
- Required fields are minimized.
- Long forms are divided into meaningful sections and may save progress.

14. Lists, Tables, and Data Density

14.1 Lists

Lists support touch-first browsing and one-item selection. Each row has a clear primary target and limited trailing actions.

14.2 Tables

Tables are preferred on desktop and some tablet views for comparison, review, and bulk import.

Table requirements:

- Persistent column meaning.
- Keyboard navigation.
- Sort and filter state visibility.
- Accessible row and column headers.
- Horizontal scrolling only when structure cannot responsibly adapt.
- Amounts aligned with tabular numerals.
- Status represented by text plus optional icon.
- Batch actions apply only to explicit selection.

14.3 Density Modes

Version 1 may support Comfortable and Compact density on desktop. Smartphone remains optimized for touch and does not expose dense mode.

15. Navigation Components

Smartphone

- Bottom navigation for Home, Transactions, Review, Search, and Settings where platform conventions allow. Assistant is optional and non-primary in Finance V1.
- Capture remains globally prominent.
- Full-screen navigation is reserved for focused tasks.

Tablet

- Adaptive sidebar or split navigation.
- Preserve selected item when transitioning between compact and expanded layouts.

Desktop

- Persistent sidebar.

- Global toolbar for Search, Capture/Import, and status.
- Secondary navigation within Records, Settings, and other large areas.
- Keyboard shortcuts and command discovery.

Navigation rules:

- Stable destinations do not move based on usage.
- AI does not replace navigation.
- Badges show actionable counts, not engagement metrics.
- Back behavior returns to the prior logical state with filters and scroll preserved.

16. Surface and Container Patterns

16.1 Card

Use for a coherent decision, summary, or preview. Avoid wrapping every piece of content in a card.

16.2 Section

Use headings and spacing to group content without unnecessary boundaries.

16.3 Sheet

Use for short choices, contextual consent, quick edit, or action preview on compact screens.

16.4 Dialog

Use for focused confirmation or authentication when leaving the current context would be harmful.

16.5 Popover/Menu

Use for compact secondary actions. Critical actions must not be hidden in ambiguous overflow menus.

16.6 Pane

Use on tablet and desktop to preserve context between lists, details, sources, and explanations.

17. Status and Feedback Components

Required status components:

- Success message.
- Warning banner.
- Error banner.
- Offline indicator.
- Synchronization delay indicator.
- Processing status.
- Security alert.
- Empty state.
- Progress indicator.
- Undo banner.

Rules:

- Success messages are brief and disappear when no longer useful.
- Warnings explain impact and next action.
- Errors state what failed, what was preserved, and recovery.
- Offline status is quiet when local work remains available.
- Security alerts are persistent until understood or resolved.
- Toasts are not used for information that must be reviewed later.

18. AI Components

18.1 AI Proposal

Displays proposed value, evidence summary, confidence when material, Accept, Edit, or Dismiss.

18.2 AI Explanation

Displays conclusion, evidence, uncertainty, action taken or recommended, and correction path.

18.3 Review Card

Displays one meaningful exception with recommendation, evidence, confidence, and one primary decision.

18.4 Assistant Answer

Displays direct answer first, supporting evidence, uncertainty, related records, and next action.

18.5 Processing Boundary

Displays whether processing occurs on-device or externally and invokes contextual consent when required.

18.6 AI Visual Rules

- AI content must not look more authoritative than user-entered or source-backed content.
- Use a subtle semantic indicator, label, or provenance line rather than decorative effects.
- Corrected user values take precedence visually and logically.
- Confidence is textual and accessible.
- AI errors use normal error patterns, not special theatrical treatment.

19. Privacy and Security Components

Required patterns:

- Privacy summary.
- External processing consent.
- Trusted-device row.
- Device fingerprint display.
- Authentication prompt.
- Security event banner.
- Data-location detail.

- Export scope selector.
- Destructive confirmation.

Rules:

- Privacy language is specific and understandable.
- Consent has no preselected approval.
- Authentication prompts explain the protected action.
- Security information is not hidden behind marketing language.
- Device revocation clearly identifies the device and confirms other trusted devices remain active.

20. Synchronization and Continuity Components

- Quiet synced state.
- Pending changes.
- Sync delayed.
- Initial device sync.
- Merge completed.
- Same-field timestamp resolution.
- Unresolved conflict.
- Revoked-device rejection.

The normal synchronized state should not occupy permanent visual space. Details remain available in My Devices or Status Center.

Different records synchronize independently. Different fields of the same record merge automatically. When the same field of the same record is changed on different devices, the latest valid timestamp determines the current value and the superseded value remains available in history when meaningful.

21. Empty States

Every empty state must answer:

- What this area is.
- Why it is empty.
- What the user can do next.

Empty states must not pressure the user, shame inactivity, or use exaggerated illustrations. Sample data may be offered where it helps understanding without mixing with real data.

22. Error States

Every recoverable error must state:

- What failed.
- What was preserved.
- Whether data left the device.
- What remains usable.

- What the user can do next.
- Whether retrying is safe.

Use inline errors for local field problems, banners for area-level issues, and full-screen error states only when the destination cannot function.

23. Content Design

23.1 Voice

Butlerly is calm, direct, respectful, specific, and modest about uncertainty.

23.2 Terminology

Use consistent product terms:

- Home
- Inbox
- Records
- Assistant
- Settings
- Capture
- Import
- Review
- Source
- Related Records
- Version History
- My Devices
- Privacy & Data

23.3 Writing Rules

- Lead with the user's situation or result.
- Use active voice.
- Use short sentences for actions and errors.
- Avoid technical architecture terms in primary UI.
- Avoid "AI-powered" marketing language inside task flows.
- Use "Butlerly" for product behavior and "Assistant" for the conversational surface.
- Do not imply emotion, consciousness, or independent authority.
- Do not blame the user.

24. Localization and Internationalization

- All text supports expansion.
- Amounts, dates, times, currencies, decimal separators, and number grouping follow locale.
- Layout supports right-to-left languages where the platform supports them.
- Icons with directional meaning mirror appropriately.
- Do not embed text in images.
- Avoid concatenated strings that prevent correct grammar.
- Financial terms use a translation glossary and product terminology review.

25. Accessibility Foundations

Every component must support:

- Screen-reader names, roles, values, hints, and state changes.
- Keyboard and switch-control completion paths where applicable.
- Visible focus.
- Large-text reflow.
- Non-color status communication.
- Reduced motion.
- Sufficient contrast.
- Touch target requirements.
- Accessible error identification.
- Logical reading order.
- Voice input compatibility where supported.

Accessibility behavior is part of component definition and acceptance, not a post-implementation annotation.

26. Platform Adaptation

26.1 Smartphone

- One-handed priority actions.
- Single-column task progression.
- Bottom navigation and prominent Capture.
- Sheets for short decisions.
- Full-screen camera and focused review.

26.2 Tablet

- Sidebar and list-detail patterns.
- Side-by-side source and record.
- Touch, pointer, keyboard, and drag-and-drop.
- Adaptive compact and expanded layouts.

26.3 Desktop

- Persistent sidebar and toolbar.
- Multi-pane workspaces.
- Keyboard-first navigation.
- Tables, bulk import, multi-select, export, and diagnostics.
- Window and panel restoration.

A platform variation may use a native component while preserving the same semantic token, state model, terminology, and outcome.

27. Theming

Version 1 supports:

- System theme.
- Light theme.
- Dark theme.

Theme requirements:

- Semantic colors are defined separately for each theme.
- Images, charts, shadows, separators, and selected states are reviewed in each theme.
- User data does not flash in the wrong theme during launch.
- Theme choice synchronizes only if the user enables preference synchronization.
- High-contrast platform settings are respected.

28. Design Assets and Documentation

Each production component requires:

- Name and unique component ID.
- Purpose.
- Anatomy.
- Variants.
- States.
- Behavior.
- Content guidance.
- Accessibility guidance.
- Platform adaptations.
- Token usage.
- Usage examples.
- Misuse examples.
- Design file reference.
- Engineering implementation reference.
- Test coverage reference.

29. Component Naming

Component names describe purpose, not appearance.

Preferred:

- Review Card
- Sync Status
- Source Preview
- AI Explanation
- Device Row
- Undo Banner

Avoid:

- Blue Card
- Rounded Box
- Magic Panel

- Special Button

IDs use C-CATEGORY-NNN, for example:

- C-ACT-001 Primary Button
- C-DATA-003 Record Row
- C-AI-002 AI Explanation
- C-SYS-004 Offline Banner

30. Governance

30.1 Ownership

Design owns semantic intent, visual behavior, content guidance, and accessibility requirements. Engineering owns implementation quality and platform integration. Product owns scope and prioritization. Changes affecting privacy, security, AI autonomy, or accessibility require cross-functional review.

30.2 Change Process

A design-system change must include:

- Problem statement.
- Affected components and screens.
- Proposed token or component change.
- Platform impact.
- Accessibility impact.
- Migration plan.
- Test plan.
- Version decision.

30.3 Versioning

- Patch: visual or documentation correction without behavior change.
- Minor: backward-compatible token, state, or component addition.
- Major: breaking semantic, API, interaction, or token change.

30.4 Exceptions

Product teams may not create one-off visual or interaction patterns without documenting why the design system cannot support the need. Valid exceptions must be reviewed and either retired or incorporated into the system.

31. Design QA

Design QA must verify:

- Correct components and tokens.
- Layout at supported breakpoints and window sizes.
- Light, dark, and system themes.
- Large text and localization expansion.
- Keyboard, pointer, touch, and screen-reader behavior.
- Loading, empty, offline, degraded, success, and error states.

- AI confidence, explanation, correction, and consent.
- Synchronization and recovery states.
- Destructive-action confirmation and Undo/history.
- No private content in analytics or notification previews by default.

32. Initial Version 1 Component Backlog

P0 Foundations

- Semantic color tokens.
- Type roles.
- Spacing and layout tokens.
- Radius, border, elevation, icon, and motion tokens.
- Light and dark themes.

P0 Controls

- Primary, secondary, tertiary, destructive, and icon buttons.
- Text, amount, date, search, select, toggle, checkbox, and radio controls.
- Menus, disclosure controls, progress, and authentication prompt.

P0 Product Components

- App shells.
- Navigation items.
- Record row and detail section.
- Capture controls.
- Import progress and summary.
- Review Card.
- AI Proposal and AI Explanation.
- Source Preview.
- Consent Sheet.
- Device Row.
- Sync Status.
- Offline/Error Banner.
- Undo Banner.
- Empty State.

P1

- Data table.
- Duplicate comparison.
- Version comparison.
- Advanced filter panel.
- Batch-action toolbar.
- QR pairing components.
- Manual transfer components.
- Diagnostics views.

33. Dependencies and Governance

UX-0007 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- UX-0005 — Butlerly Screen and State Inventory.
- UX-0006 — Butlerly AI Interaction and Explainability Specification.
- Product MVP scope and personas.
- Architecture decisions for supported platforms, local storage, synchronization, AI providers, security, and accessibility.

UX-0007 governs:

- UX-0008 — Cross-Device Experience and Continuity.
- UX-0009 — Accessibility Standard.
- UX-0010 — Low-Fidelity Wireframe Specification.
- UX-0011 — Design Patterns and Component Library.
- Prototypes, engineering UI implementation, design QA, and release acceptance.

34. Acceptance Criteria

This document is complete when:

- Semantic foundations are defined for color, type, spacing, layout, shape, elevation, iconography, and motion.
- Component architecture and naming are clear.
- Controls, status, navigation, AI, privacy, synchronization, error, and recovery patterns are defined.
- Smartphone, tablet, and desktop adaptations are explicit.
- Light, dark, system, accessibility, localization, and reduced-motion behavior are covered.
- Designers can create consistent wireframes and high-fidelity designs without inventing new foundations.
- Engineers can implement tokens and reusable components with testable states.
- QA can verify visual, interaction, privacy, AI, synchronization, and accessibility behavior.
- Governance prevents uncontrolled one-off patterns and supports versioned evolution.

35. BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 is the implementation authority for visual tokens, concrete color values, typography sizes, spacing/radius values, responsive breakpoints, theming, component contracts, Flutter design-system implementation, localization/original-data presentation, and design QA. Where this foundation document is more general or uses an older example, BDS-0000 governs implementation.

Finance V1 stable primary navigation is Home, Transactions, Review, Search, and Settings. Assistant remains optional and non-primary. Legacy Inbox/Records wording in explanatory examples must not be interpreted as permission to change the current visible primary

navigation. Core operation requires no Butlerly account; device trust and synchronization patterns are capability-driven and future/optional unless product scope explicitly enables them. UI language follows the selected application language, while user-entered/imported/scanned/source data preserves its original language by default; original currency/amount remains authoritative and any reference-currency conversion is shown separately.